INCH-POUND

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22 June 1990
SUPERSEDING
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15 November 1977

MILITARY SPECIFICATION

PROPELLANT M9 FOR USE IN CARTRIDGE, IGNITION M702

This specification is approved for use within the US Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers M9 propellant for use in Ignition Cartridge M702 for 60MM mortars.

1.2 Classification. The propellant shall be M9 Flake.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and Supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander, U.S. Army, ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1376

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SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For

STANDARDS

MILITARY

MIL-STD-652 - Propellants, Solid, for Cannons, Requirements and Packing

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other government documents, drawings, and publications. The following other government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified the issues shall be those in effect on the date of the solicitation.

DRAWINGS

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

9280553 - Cartridge, Ignition, M702 Loading Assembly

(Copies of specifications, standards, other government documents, drawings and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the reference cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. The propellant shall be M9 Flake, conforming to the requirements of MIL-STD-652 and with a nominal thickness of .006 inches.

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3.2 Ballistic assessment requirements. The propellant when loaded into the cartridge, ignition, M702 and with the weight of projectile M720, 3.75 ± .01, pounds (lbs) for which it was manufactured and drop fired in the M224 mortar, shall comply with the following requirements when tested at 70 degrees F and standard weapon condition (see 6.9) at charge 0. The propellant shall be capable of assessment to a velocity of 210 feet per second (fps) with an average pressure not to exceed 1650 pounds per square inches (psi).

3.3 Ballistic uniformity requirements. The propellant when fired in the M224 mortar at recommended charge weight and with the projectile M720 for which it was manufactured, shall comply with the following requirements when tested as specified in 4.5.1.2.

3.3.1 Uniformity of velocity. The standard deviation of velocity for the lot at 70 degrees F shall not exceed 3.5 fps.

3.3.2 Average pressure. The propellant shall not produce an average pressure greater than 1650 pounds per square inch (psi) at 70 degrees F and 1940 psi at 145 degrees F.

3.3.3 Individual maximum pressure. At 145 degrees F, the propellant shall not produce an individual pressure greater than 2670 psi or an average pressure plus three (3) standard deviation greater than 2670 psi.

3.4 Preproduction inspection. This specification contains provisions for preproduction inspection. Requirements for the submission of preproduction inspection samples by the contractor shall be as specified in the contract.

3.5 Workmanship. The best manufacturing practices shall be used in the formulation of propellant furnished under this specification, and all other applicable documents. The propellant and its standard ingredients shall be protected from the action of direct sunlight and acid fumes. Unless otherwise specified, no reground propellant or nitrocellulose shall be used in the manufacture of propellant procured under this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements

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(examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. Preproduction sample inspection and Quality Conformance Inspection shall be conducted on this item.

4.3 Preproduction sample.

4.3.1 Submission. The contractor shall submit a preproduction sample as designated by the Contracting Officer for evaluation in accordance with 4.3.2. The preproduction samples shall consist of up to three (3) pilot lots, approximately 50 lbs each and shall expect to bracket or meet the requirement of this specification. A new sample, or portion thereof, as directed by the contracting officer, shall also be submitted whenever a change occurs in manufacturing process, material used, drawing, specification or source of supply as to significantly affect product uniformity, loadability, or ballistic performance as determined by the Technical Agency (see 6.11).

4.3.2 Inspection to be Performed. Each of the pilot lots comprising the preproduction sample and submitted in accordance with 4.3.1 may be subjected by the Government to any or all of the requirements of the applicable drawings and specifications.

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4.3.3 Rejection. Government engineering, Product Assurance personnel and the contractor shall review the pilot lots test data and shall select the granulation which best fits the requirements of this specification. The technical agency shall notify the Contracting Officer of the selected granulation. The Contracting Officer shall designate the chosen pilot lots as the accepted preproduction sample and shall authorize the contractor to produce the selected granulation. If none of the pilot lot submitted meets or brackets the requirements of this specification, a new preproduction sample shall be submitted in accordance with 4.3.1. Until the preproduction sample is accepted, the contractor is in no way authorized by the Government to initiate regular production, unless otherwise directed by the Contracting Officer.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

PRECAUTION: This specification covers sampling and testing of chemical or explosive materials which are potentially hazardous to personnel. Accordingly, it is emphasized that all applicable safety rules, regulations and procedures shall be followed in handling and processing these materials.

4.4.2 Examination and testing.

4.4.2.1 Sampling and testing for chemical and physical characteristics. Each lot of propellant shall be sampled and tested for composition, form dimensions and chemical/physical properties in accordance with MIL-STD-652.

4.4.2.2 Ballistic assessment. Representative samples shall be taken by random selection of ten containers from each propellant lot. Equivalent quantities of the ten containers shall be taken such that the aggregate quantity shall be sufficient for the ballistic tests described in 4.5.1. Each container's sample shall be packaged separately, sealed, and marked to show propellant lot number and container serial number. The ten samples shall be shipped to a designated LAP facility where the propellant will be loaded in accordance with drawing 9280553.

4.4.2.3 Velocity uniformity (70 degrees F) (see 3.3.1). If the product of the factor 0.73 times the velocity standard deviation of the rounds tested in accordance with 4.5.1.2 exceeds

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the applicable requirement, apply retest and referral criteria in accordance with 4.5.1.3 and 4.5.1.4. If the retest and referral criteria do not apply, the lot shall be rejected.

4.4.2.4 Average pressure (see 3.3.2). If any average pressure obtained during the uniformity testing exceeds the applicable requirements, apply retest and referral criteria in accordance with 4.5.1.3 and 4.5.1.4. If the reject and referral criteria do not apply the lot shall be rejected.

4.4.2.5 Individual maximum pressure (see 3.3.3). If any individual pressure observed during the 145 degrees F uniformity series or the average pressure plus three times the standard deviation exceeds the applicable requirement, testing shall cease and the lot shall be rejected.

4.5 Methods of inspection.

4.5.1 Ballistic assessments. These ballistic tests shall be conducted at a Government Proving Ground (designated by the contracting officer) in accordance with the applicable Proving Ground Acceptance Test Procedure. All rounds shall be loaded and inspected by a designated LAP facility in accordance with Drawing 9280553 (or appropriate alternate so specified by the contracting officer). Rounds shall be temperature conditioned in a large conditioning box, with adequate air circulation to assure uniform conditioning, at the required temperature for a minimum of 24 hours and assurance will be made that, when fired, the rounds are at the conditioning temperature. The following tolerances shall apply to all temperature conditioning: -50 degrees F, + 5 degrees F; 70 degrees F, $\pm$ 2.5 degrees F; 145 degrees F, -5 degrees F. Copper crusher gages (2) shall be used to measure the maximum chamber pressure of each round. Velocity coils will be used to measure the muzzle velocity of each round.

4.5.1.1 Charge establishment. Three groups of three rounds each, loaded with three selected charge weights of propellant (weights selected should bracket expected service charge weight) shall be conditioned at 70 degrees F, and drop fired in the M224 mortar at an elevation of 45 degrees. Each round shall be loaded from a separate sample container. Each round shall be marked with the propellant lot number, sample container, serial number and charge weight. From the data obtained, a charge weight versus pressure versus velocity curve shall be established. The charge weight which will yield the prescribed service velocity shall be calculated (see 3.2). This established charge weight of propellant shall then be loaded into the rounds for the uniformity series (see 4.5.1.2).

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4.5.1.2 Uniformity. Thirty rounds shall be loaded with the established charge weight of propellant. Three groups of ten rounds each shall be conditioned to -50 degrees F, 70 degrees F, and 145 degrees F, respectively, and drop fired in the M224 mortar at an elevation of 45 degrees. Within each ten round group, the individual rounds shall be loaded with propellant withdrawn from separate sample containers as described in 4.4.2.2. Each round shall be marked with the propellant lot number, sample container serial number, and charge weight. Rounds fired at 70 degrees F shall be fired alternately with calibration rounds and shall be corrected to standard weapon/firing conditions (i.e., the difference between calibration performance under test conditions and standard conditions). Firing results at all temperatures shall be corrected to standard projectile weight (see 3.2). The means and standard deviations of the velocities and pressures at all firing temperatures shall be calculated using corrected results. The difference between the corrected velocity obtained during the uniformity series and the service velocity shall be compensated by a final adjustment in the charge weight using the slope of the velocity charge weight curve established in 4.5.1.1. The resultant will be the recommended charge weight.

4.5.1.3 Retest criteria. A lot shall be retested whenever any of the following conditions exist and whenever the proving ground deems it necessary (see 6.7).

a. If the final adjustment in charge weight exceeds two percent of the charge weight used in uniformity firings, repeat the charge establishment and uniformity at 70 degrees F.

b. If for any reason, the proving ground considers that test conditions have detrimentally affected the test results, additional rounds as required shall be tested.

4.5.1.4 Referral criteria. A test lot shall be referred to the technical agency for disposition whenever any of the following conditions exist. (See 6.2.2 for addresses of the technical agency.)

a. Calibration rounds fired in any test phase fail to comply with Section 3 of this specification.

b. Calibration correction applied exceeds two percent of the expected muzzle velocity.

c. Calibration components are not available.

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d. Number of test or calibration results utilized in calculations is less than specified.

e. Any unusual occurrences during ballistic testing, such as excessive delays between rounds, stickers, or projectile flights, etc. Each such event shall be reported in detail on the applicable firing record.

f. Test lot fails velocity standard deviation or average pressure requirement at 70 degrees F and any extreme test value is an outlier.

The following test shall be performed to determine if a single test result is an outlier with a sample size of 10 at a significance level of 0.05, the critical value (T_c) is 0.477. Calculate a test value (T_t) as follows:

For Low Outlier

$$T_t = \frac{(X_2 - X_1)}{(X_9 - X_1)}$$

For High Outlier

$$T_t = \frac{(X_{10} - X_9)}{(X_{10} - X_2)}$$

Where:

X_1 = test result being tested as low outlier
X_2 = test result adjacent to X_1, after all the test results are ordered from smallest to largest
X_9 = tests result adjacent to the largest test result
X_{10} = test result being tested as high outlier

X_1 or X_{10} is an outlier if T_t is greater than T_c .

5. PACKAGING

5.1 Packing and marking. Packing and marking shall be in accordance with MIL-STD-652, except that all propellant designated for use as reference shall be packed Level A.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This M9 propellant is intended for use in the M702 ignition cartridge for 60mm mortars.

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6.2 Ordering data. Ordering data shall be in accordance with MIL-A-48078 (PA) with the following exceptions.

6.2.1 Acquisition requirements. Acquisition requirements of MIL-A-48078 (PA) shall apply.

6.2.2 Contract data requirements. In addition to the normal distribution of records, when the propellant is procured by the Department of the Army, one copy of all ballistic test data and one copy of the Propellant Description sheet shall be forwarded to: Commander, U.S. Army Armament, Munitions and Chemical Command, ATTN: AMSMC-QAR-R(D), and one copy to ATTN: SMCAR-AEE-DP(D), Picatinny Arsenal, NJ 07806-5000.

6.3 Standard deviation. The standard deviation shall be calculated with (n-1) as the divisor in a standard statistical technique equivalent to that shown in MIL-STD-414, Section B, example B-1.

6.4 Surveillance sample. Samples shall be forwarded to U.S. Army Armament, Munitions and Chemical Command, Picatinny Arsenal, NJ, ATTN: SMCAR-AEE-WE(D) for the propellant surveillance test program in accordance with MIL-STD-652.

6.5 Test rounds. Test rounds shall be assembled in accordance with drawing 9280553.

6.5.1 Test cartridges loaded at a load plant. Sample quantities of M9 propellant shall be selected for ballistic tests in accordance with MIL-STD-652 except that selection shall be from ten propellant containers. Each of the samples shall be packaged separately, sealed and marked to show the propellant lot number and container serial number from which the propellant was withdrawn. The ten sample containers shall be shipped to a designated loading facility where the propellant will be assembled into M702 ignition cartridges. Upon arrival at the facility, the sample containers shall be moved to an explosive holding area. Prior to unsealing, each sample container shall be rolled and tumbled (to mix the contents) and set aside for 24 hours (until temperature of the contents has stabilized with that in the holding area). The samples shall be assembled into M702 ignition cartridges in accordance with drawing 9280553. Nine test cartridges shall be assembled for the charge establishment tests using charge weights of 50, 52, and 54 grains, i.e. three cartridges per charge weight and representing one cartridge per propellant container. Each test cartridge shall be marked with the propellant lot number, propellant container serial number and charge weight. Thirty test cartridges shall be subsequently

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assembled for the uniformity test using the charge weight obtained in the charge establishment tests. Three test cartridges shall be assembled per propellant container. Each test cartridges shall be marked with the propellant lot number, propellant container serial number and charge weight. One cartridge per propellant container shall be tested at each temperature level of the uniformity tests.

6.6 Proving ground test summary.

Temp (deg F)	Sample Size	Assessed Velocity (fps)	Lot Std Dev of Vel (fps) max	Ind Max Press (psi)	Avg Press Max (psi)
-50	10	N/A	N/A	N/A	N/A
70	10	210	3.5	N/A	1650
145	10	N/A	N/A	2670	1940

6.7 Retest. All original and retest data shall be submitted to U.S. Army Armament, Munitions and Chemical Command, ATTN: AMSMC-QAR-R(D), Picatinny Arsenal, NJ 07806-5000.

6.8 Test anomalies. Any unusual occurrences during ballistic testing, such as excessive delays between rounds, stickers, poor projectile flights, etc., shall be reported in detail on the applicable firing record.

6.9 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament, Munitions and Chemical Command (AMCCOM) may also include drawings prepared by, and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of AMCCOM.

6.10 Standard weapon. Standard weapon and standard weapon conditions refer to the weapon model and conditions of the test at the time the master calibration series was fired.

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6.11 Preproduction sample. The preproduction sample is intended to provide the contractor with the necessary latitude to establish the propellant granulation needed to meet the required ballistic parameters. The Technical Agency and the contractor should jointly select the one pilot lot which best represents the desired parameters. All future production lots should be of the same basic granulation as the accepted pilot lot.

6.12 Reference sample. Whenever a propellant lot is selected for reference use, a 5- lb sample shall be forwarded to US Army Armament, Munitions and Chemical Command, ATTN: SMCAR-AEE-WE(D), Picatinny Arsenal, NJ 07806-5000, for use in evaluating the propellant stockpile.

6.13 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE *(YYMMDD)*
3. DOCUMENT TITLE		
4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*		
5. REASON FOR RECOMMENDATION		

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a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
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a. NAME	b. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON
c. ADDRESS *(Include Zip Code)*	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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